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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

SECURE USER LOGIN AND AUTHENTICATION SYSTEM WITH PASSWORD VALIDATION

NAME : NETHRA.A
REGISTER NUMBER: 25BCS241
COURSE NAME : B.Sc.CS
DEPARTMENT : B.Sc.CS-I-C
SUBJECT :JAVA PROGRAMMING

INTRODUCTION

1. Secure login systems are essential in modern software applications.
2. They protect user data from unauthorized access.
3. Authentication ensures only valid users can access the system.
4. Password validation improves security by enforcing strong password rules.

PROBLEM STATEMENT:

1. Existing manual or weak login systems are insecure.
2. Users may create weak passwords that are easy to guess.
3. Unauthorized access can lead to data theft or misuse.
4. There is a need for an automated and secure authentication system.

OBJECTIVES OF THE PROJECT

1. Develop a secure user login system.
2. Implement password validation rules.
3. Reduce security risks and unauthorized access.
4. Improve reliability and efficiency of user authentication.

SCOPE OF THE SYSTEM

The system can be used in:

1. Educational institutions
2. Online portals
3. Banking and financial applications
4. Training institutes
5. Small business management systems

SYSTEM MODULES

Module	Description
Login	Verify username and password
Register	Create new user account
Password Validation	Check password strength
Update Password	Modify existing password
Logout	Securely exit the system

SYSTEM ARCHITECTURE /WORKFLOW

Start

↓

User Registration / Login

↓

Password Validation

↓

Authentication Success / Failure

↓

Access System Modules

↓

Logout

↓

End

JAVA CONCEPTS USED

1.Classes and Object : Used to create the main login system class and GUI components.
2.Constructor:Used to initialize the login window and components when the program starts
3.Inheritance:The main class extends JFrame to use GUI features.
4.Event Handling:ActionListener is used to handle Register and Login button clicks.
5.Exception Handling:Used to handle possible runtime errors like invalid input.
6.String Handling:Used for username comparison and password validation.
7.Password Validation Logic:Conditions like minimum length, uppercase letter, and digit checking.
8.Java Swing (GUI):Components like JFrame, JTextField, JPasswordField, JButton, JOptionPane.
9File Handling (Optional Enhancement):Used to store user credentials permanently.

JAVA PROGRAM

Main.java

```java
import java.util.Scanner;

class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Stored credentials
        String correctUsername = "admin";
        String correctPassword = "Admin@123";

        // User input
        System.out.print("Enter Username: ");
        String username = sc.nextLine();

        System.out.print("Enter Password: ");
        String password = sc.nextLine();

        // Password Validation (simple rules)
        if (password.length() < 8) {
            System.out.println("Password must be at least 8 characters");
        } else if (!password.matches(".*[A-Z].*")) {
            System.out.println("Password must contain at least one uppercase letter");
        } else if (!password.matches(".*\\d.*")) {
            System.out.println("Password must contain at least one number");
        } else {
            // Authentication
            if (username.equals(correctUsername) && password.equals(correctPassword)) {
                System.out.println("Login Successful");
            } else {
                System.out.println("Invalid Username or Password");
            }
        }

        sc.close();
    }
}
```

input

OUTPUT SCREEN

```
Enter Username: admin
Enter Password: admin@123
Password must contain at least one uppercase letter


...Program finished with exit code 0
Press ENTER to exit console.
```

DEMONSTRATION STEPS

1. Run the application.
2. Register a new user account.
3. Enter username and password.
4. System validates password strength.
5. If correct → access granted.
6. Perform operations and logout.

CONCLUSION

1. The project successfully implements a secure login system.
2.Password validation improves overall application
3.security Demonstrates practical use of Core Java programming concepts.
4.Provides a foundation for developing real-world secure systems.

FUTURE ENHANCEMENTS

1. Database integration (MySQL / Oracle)
2. OTP or Two-Factor Authenticatio
3. Web-based login system
4. User role management (Admin / User)
5. Password encryption using hashing algorithms

REFERENCE

The following resources were used during the development of this project:
Books
Java: The Complete Reference
Websites
Oracle – https://www.oracle.com/java
GeeksforGeeks – https://www.geeksforgeeks.org
W3Schools – https://www.w3schools.com
Javatpoint – https://www.javatpoint.com